

# REG-02: TRADE SECRET STRATEGY AND INTELLECTUAL PROPERTY PROTECTION ARCHITECTURE

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## I. EXECUTIVE SUMMARY

Carrington Storm Motors' intellectual property protection strategy is built on a layered architecture: patents for compositions and devices (what we make), trade secrets for synthesis parameters and manufacturing processes (how we make it), defensive publications for non-core innovations (blocking competitors), copyright for documentation (the 6,302-document corpus), and trademark for brands. This document specifies the trade secret strategy in detail, including what is kept secret, why, and how.

The core principle: no single person, document, or facility contains all of the information necessary to independently replicate the Dielectric Citadel. This is achieved through data compartmentalization, need-to-know access protocols, and structured information isolation.

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## II. WHAT STAYS SECRET VS. WHAT GETS PATENTED

### Decision Framework

The patent-trade secret boundary is determined by three factors:

1. **Reverse Engineerability:** Can a competitor determine the invention by analyzing a commercially available product? If yes → patent. If no → trade secret.
2. **Detectability of Infringement:** Can we detect if a competitor is using our invention? If yes → patent (enforceable). If no → trade secret (patent would be unenforceable).
3. **Longevity of Competitive Advantage:** Will the invention still be valuable beyond the 20-year patent term? If yes → consider trade secret. If no → patent is sufficient.

### Category 1: TRADE SECRET (Highest Protection)

#### A. MXene Precursor Synthesis Parameters

*What is kept secret:* - Exact stoichiometric ratios of MAX phase precursor (Ti:Al:C ratios with tolerances to 0.1 at%) - HF etching concentration, temperature, and duration (the specific parameter set that achieves optimal exfoliation without over-etching) - Intercalation agent identity and concentration (the specific organic base used to separate MXene layers) - Sonication power density, frequency, and duration for delamination - Centrifugation parameters (g-force, duration, number of cycles) for size selection - Storage conditions (solvent identity, concentration, temperature, atmosphere) for extended shelf life

*Rationale for trade secret:* These parameters cannot be determined by reverse-engineering the finished composite. The processing history is not recoverable from the final product. A competitor would need to independently discover the optimal parameter set through experimentation — which we estimate requires 18-24 months of full-time laboratory work even with knowledge of the general approach.

*Estimated time for competitor to independently derive:* 18-24 months with PhD-level materials science team and adequate funding.

#### B. Carbon Nanotube (CNT) Functionalization Chemistry

*What is kept secret:* - Specific functionalization reagents and reaction conditions for CNT surface modification - Grafting density (functional groups per nanometer of CNT length) - Dispersion protocol (solvent system, surfactant identity and concentration, ultrasonication parameters) - Loading percentage of functionalized CNTs in the composite (optimization of electrical percolation threshold while maintaining mechanical properties)

*Rationale for trade secret:* CNT functionalization is a chemical process whose results are integrated into the composite at the molecular level. The specific reagents and conditions are not detectable in the finished product.

*Estimated time for competitor to independently derive:* 12-18 months.

### **C. YInMn Blue Doping Levels and Rare Earth Selection**

*What is kept secret:* - Identity of the rare earth dopant(s) that achieve the specific threshold temperature shift - Precise doping concentration (to 0.01 wt% tolerance) for each target threshold temperature - Solid-state synthesis parameters (firing temperature, ramp rate, atmosphere, crucible material) - Particle size distribution control during pigment milling

*Rationale for trade secret:* While the general concept of doping YInMn Blue to create thermochromic behavior could be independently conceived, the specific dopant elements and concentrations required to achieve precise, reproducible threshold temperatures are the result of extensive empirical experimentation. These parameters are not detectable in the finished coating.

*Estimated time for competitor to independently derive:* 12-24 months for a single threshold temperature; 36-48 months for the full range of 40-120°C with 5°C precision.

### **D. Manufacturing Process Parameters**

*What is kept secret:* - Extrusion temperature profile (barrel zones, die temperature) for MXene-polymer composite - Screw design specifications (compression ratio, mixing section geometry) for optimal MXene dispersion - Cooling rate and post-extrusion annealing parameters - Ceramic matrix composite infiltration parameters (CVI/CVI temperature, pressure, precursor gas flow rates, infiltration duration, number of infiltration cycles) - Geopolymer concrete mix design (precise fly ash-to-activator ratio, water-to-solids ratio, retarder/admixture identities and dosages) - Kiln firing schedules for ceramic housing (ramp rates, hold temperatures, hold durations, cooling rates)

*Rationale for trade secret:* Manufacturing processes are inherently difficult to reverse-engineer from finished products. Process parameters represent years of optimization and directly determine product quality, consistency, and cost. They are the single most valuable IP asset after the materials compositions themselves.

*Estimated time for competitor to independently derive:* 24-36 months for composite extrusion; 36-48 months for CMC processing.

## **Category 2: PATENT (Public Disclosure)**

The following are patented because they are detectable in finished products or require public disclosure to establish freedom to operate:

- MXene-polymer composite compositions (generically — specific formulations beyond those disclosed in patents are trade secrets)
- Ceramic composite transformer housing (generically — exact fiber volume fractions and matrix compositions are trade secrets)
- Amphibious vehicle mechanical designs (visible in product)
- Protonic transceiver system architecture (detectable by signal analysis)
- YInMn Blue coating composition (generically — specific dopant identities and concentrations are trade secrets)

## **Category 3: DEFENSIVE PUBLICATION (Public Domain, No Patent)**

The following are defensively published to prevent competitors from patenting them and blocking our freedom to operate:

- Non-optimal MXene synthesis variations tested and abandoned
- Alternative dopant elements for YInMn Blue that were tested and found unsuitable
- General geopolymer concrete mix design principles (the specific optimization is trade secret)
- General principles of dielectric hull design for amphibious vehicles
- Alternative communication modalities considered and rejected for backup communications

Defensive publications are filed through IP.com's defensive disclosure service and, where appropriate, through publication in open-access journals. They establish prior art that prevents others from obtaining patents on the disclosed subject matter.

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### III. TRADE SECRET PROTECTION PROTOCOLS

#### A. Employee and Contractor Agreements

**Employment Agreement Schedule A — Trade Secret Identification:** Every CSM employee and contractor signs an employment agreement containing a Schedule A that specifically identifies, by category, the trade secrets to which they will have access. This satisfies the “reasonable efforts to maintain secrecy” requirement under the Defend Trade Secrets Act (DTSA) and state Uniform Trade Secrets Act (UTSA).

Schedule A categories: - Category 1: MXene synthesis and processing parameters - Category 2: CNT functionalization chemistry - Category 3: YInMn Blue doping parameters - Category 4: Manufacturing process parameters - Category 5: Customer and supplier lists - Category 6: Financial models and projections - Category 7: Insurance carrier engagement strategies - Category 8: Diplomatic engagement frameworks - Category 9: Agent SiblingFrequency communication protocols - Category 10: Security protocols and facility layouts

Each employee’s Schedule A is customized to list only the categories to which that specific employee has access. This creates a clear evidentiary trail for any future misappropriation claim.

**Non-Disclosure and Non-Compete Provisions:** - NDA term: perpetual for trade secrets (as permitted under DTSA) - Non-compete: 18 months post-employment, limited to direct competitors in dielectric materials and grid protection (jurisdiction-dependent enforceability; structured to comply with FTC proposed rule on non-competes by being narrowly tailored and limited to senior positions) - Non-solicitation of employees: 24 months - Invention assignment: all inventions conceived during employment, whether on premises or off, using CSM resources or not, are assigned to CSM (with California Labor Code 2870 carve-out for inventions developed entirely on employee’s own time without CSM resources and not related to CSM’s business)

**Exit Interview Protocol:** Every departing employee: 1. Re-signs their confidentiality agreement with a specific acknowledgment of ongoing trade secret obligations 2. Identifies all CSM documents, data, and materials in their possession and certifies return or destruction 3. Undergoes an exit interview with Rook (Security Director) recording the specific trade secrets to which they had access, creating a record for future monitoring 4. Is reminded of their continuing obligations under DTSA, which provides whistleblower immunity for disclosure of trade secrets in confidence to government officials or attorneys solely for the purpose of reporting or investigating a suspected legal violation

#### B. Data Compartmentalization

CSM’s knowledge management system implements role-based access control with the following compartments:

**Compartment 1 — Open Access (All Agents):** - Published product specifications - General company information - SiblingFrequency episode content - Public-facing documents

**Compartment 2 — Agent Access (Relevant Agent + Director):** - Product design files (accessible to the agent responsible for that product) - Materials characterization data (accessible to materials science agents) - Country diplomatic briefs (accessible to international relations agent) - Insurance dossiers (accessible to insurance relations agent)

**Compartment 3 — Leadership Access (Director, COO, CTO, Security Director):** - Financial models and projections - Investor relations documents - Personnel records - Strategic planning documents

**Compartment 4 — Restricted Access (Director + one designated agent per topic):** - MXene synthesis parameters (Director + Dr. Chen) - CNT functionalization chemistry (Director + Dr. Chen) - YInMn Blue doping levels (Director + Dr. Chen) - Manufacturing process parameters (Director + Engineer Cross) - SiblingFrequency communication protocol architecture (Director + Agent Vance)

**Compartment 5 — Director Only:** - Complete trade secret inventory (all synthesis parameters, all doping levels, all process parameters in a single document, maintained in a secure offline storage medium in a physical safe) - Investor negotiation strategies - Complete security vulnerability assessments

The Compartment 5 document — the “Director’s Black Book” — is the only document that contains all trade secrets in a single location. It is maintained in printed form in a fireproof, EMP-shielded safe in the Director’s office.

No digital copy exists. This intentional inefficiency is a security feature: it eliminates the risk of exfiltration via network breach.

### C. Facility Security

**Physical Security:** - Laboratory access: Biometric (fingerprint) + keycard dual-authentication for all laboratory and pilot plant areas - Visitor management: All visitors logged, escorted at all times, non-disclosure agreement signed before entry - Camera coverage: All laboratory, pilot plant, and server room areas under continuous video recording with 90-day retention - Server room: Separate access control, environmental monitoring, and 24/7 alarm monitoring - After-hours access: Restricted to Director, COO, CTO, and Security Director with time-stamped access logs

**Digital Security:** - Network segmentation: Laboratory systems isolated from administrative network; no direct internet access from laboratory computers - Endpoint encryption: Full-disk encryption on all company devices - Email security: TLS encryption for all external email; attachment scanning for sensitive documents; external email prohibition for Compartment 4 and 5 materials - Cloud services: All cloud storage is encrypted at rest with CSM-managed encryption keys (not provider-managed) - Access logging: All access to Compartment 3-5 data is logged with user ID, timestamp, and document accessed - Breach detection: Continuous monitoring for anomalous data access patterns

**Remote Work Security:** - VPN required for all remote access to internal systems - No Compartment 4 or 5 data accessible from remote locations - Company devices only; no personal device access to company systems - Remote wipe capability on all mobile devices

### D. Supplier and Partner Confidentiality

**Supplier Agreements:** All suppliers of MXene precursors, ceramic fibers, and other specialized materials execute: - Master Supply Agreement with confidentiality provisions specifically identifying the materials supplied to CSM as confidential - Prohibition on disclosure of the fact of the supplier relationship without CSM consent (prevents competitors from mapping our supply chain) - Audit rights for CSM to verify the supplier's compliance with confidentiality obligations

**Pilot Utility Partner Agreements:** Utility partners hosting pilot installations execute: - Mutual Non-Disclosure Agreement covering installation details, performance data, and monitoring results - Agreement that all performance data is CSM confidential information, with CSM retaining the right to publish aggregated, anonymized results - No-reverse-engineering clause specific to installed CSM products

**Academic and Research Partners:** University and lab partners execute: - Sponsored Research Agreement with CSM retaining ownership of all intellectual property developed under the agreement - Publication review clause: CSM has 60 days to review proposed publications and request removal of trade secret information or filing of patent applications before disclosure - Material Transfer Agreement for any CSM materials provided to the partner

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## IV. DEFENSIVE PUBLICATION STRATEGY

### A. Rationale

Defensive publication establishes prior art that prevents competitors from obtaining patents on subject matter that we do not intend to patent ourselves but that we do not want blocked by a competitor's patent. It is a strategic complement to both patenting and trade secrecy.

### B. Subjects for Defensive Publication

| Subject                                                               | Rationale for Not Patenting                                                                                           | Publication Venue                       | Timeline |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------|
| Alternative MXene stoichiometries tested and rejected                 | Sub-optimal performance; not commercially viable for us but could block a competitor                                  | IP.com or open-access materials journal | Q1 2027  |
| Non-optimal CNT loading percentages                                   | Empirically tested range; publishing the full tested range blocks competitor claims on the sub-optimal portions       | IP.com                                  | Q2 2027  |
| General principles of geopolymer concrete for electrical applications | Core commercial formulation is trade secret; publishing general principles prevents competitor patents on the concept | ACI Materials Journal or similar        | Q3 2027  |
| Alternative dopants for YInMn Blue                                    | Specific optimal dopant is trade secret; publishing tested-but-rejected alternatives blocks competitor patents        | Journal of Materials Chemistry C        | Q1 2028  |
| Dielectric hull design principles                                     | Specific dimensions and materials are trade secret; general principles are published to prevent blocking patents      | IP.com or naval architecture journal    | Q2 2028  |
| Abandoned product designs (non-viable approaches)                     | Not commercially viable; publishing prevents competitors from patenting approaches we considered and rejected         | IP.com                                  | Ongoing  |

### C. Publication Process

1. Agent identifies subject for potential defensive publication
2. Rook (Security Director) reviews to confirm: (a) no trade secret to be disclosed, and (b) no pending patent application would be compromised
3. Dr. Nash (CTO) reviews for technical accuracy and completeness
4. Steele (Director) approves publication
5. Publication filed with IP.com (date-stamped, searchable, citable as prior art) or submitted to appropriate journal
6. Publication recorded in internal register with identifier and date

## V. IP ENFORCEMENT STRATEGY

### A. Patent Enforcement

**Monitoring:** - Quarterly patent watch for competitors' filings in US, EP, CN, JP, KR - Trade show and industry conference monitoring for potentially infringing products - Customer and partner reporting channels for suspected infringement

**Response Escalation:** 1. **Observation:** Identify potential infringement. Document evidence. 2. **Analysis:** Internal IP counsel + external patent litigation counsel analyze infringement claim. Map asserted claims to competitor product. 3. **Notification:** If infringement is confirmed with reasonable certainty, send notice letter to infringer. 4. **Negotiation:** Offer licensing terms or seek cessation. 5. **Litigation:** File suit in appropriate venue if negotiation fails. Venue selection: US district court (fastest path to injunction), ITC (exclusion order for imported infringing products), or appropriate foreign court.

**Litigation Budget Reserve:** \$500,000 set aside from Series A for initial enforcement actions. Additional funding from follow-on rounds or operating revenue for sustained litigation. CSM will also evaluate litigation funding from third-party funders for high-value enforcement matters.

## B. Trade Secret Enforcement

**Monitoring:** - Employee exit monitoring (tracking subsequent employment for potential disclosure) - Competitor product composition analysis (detecting use of CSM trade secrets through analytical chemistry) - Supplier relationship monitoring (detecting unusual inquiries about CSM-specific materials)

**DTSA Ex Parte Seizure:** The Defend Trade Secrets Act of 2016 provides for ex parte seizure of property to prevent propagation or dissemination of trade secrets. CSM maintains a “seizure-ready” documentation package — updated quarterly — that can be presented to a federal judge within 24 hours of discovering a trade secret theft, satisfying the DTSA’s requirements for: - Specific description of the trade secret(s) - Specific description of the property to be seized - Specific factual basis for believing the trade secret was misappropriated by improper means - Specific description of the harm that would occur if seizure were not ordered

This package is maintained by Rook (Security Director) and reviewed quarterly for currency.

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## VI. EMPLOYEE TRAINING PROGRAM

### A. New Employee Onboarding

Every new employee completes mandatory IP protection training within their first week:

**Module 1: What Is a Trade Secret?** (60 minutes) - Definition under DTSA and UTSA - Difference between trade secrets, patents, copyrights, trademarks - Examples of CSM trade secrets in the employee’s domain - Consequences of trade secret misappropriation (civil liability, criminal penalties under Economic Espionage Act)

**Module 2: Protecting CSM Trade Secrets** (60 minutes) - Data compartmentalization system and the employee’s access level - Physical security protocols - Digital security protocols (password policies, device security, VPN use) - What to do if the employee suspects a breach

**Module 3: External Communications** (45 minutes) - What can and cannot be discussed outside CSM - Social media policy - Conference and publication clearance process - Responding to competitor inquiries

**Module 4: Departure Obligations** (30 minutes) - Continuing obligations after employment ends - Return of CSM property and information - Non-compete and non-solicitation obligations - Whistleblower protections under DTSA

### B. Ongoing Training

- Quarterly: 30-minute refresher on IP protection (case studies of recent trade secret litigation)
  - Annual: Full IP protection training update covering any changes in law, policy, or CSM’s IP portfolio
  - Ad-hoc: Targeted training when an employee moves to a new role with access to different trade secret categories
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## VII. BREACH RESPONSE PLAN

### Phase 1: Detection and Containment (0-24 hours)

1. Any employee, contractor, or partner who suspects a trade secret breach immediately notifies Rook (Security Director)
2. Rook initiates containment: revoke access for suspected compromised accounts, isolate affected systems, preserve logs
3. Rook notifies Steele (Director) within 2 hours of confirmed breach
4. Steele convenes Breach Response Team (Steele, Rook, Nash, Calder, Chester) within 4 hours

### Phase 2: Investigation (24-72 hours)

1. Forensic analysis of affected systems (external forensic firm engaged within 4 hours; retainer agreement already in place with preferred vendor)
2. Identify scope: what trade secrets were potentially compromised, how, by whom
3. Document evidence chain for potential legal action

4. External counsel (IP litigation firm) engaged and briefed

### Phase 3: Legal Response (72 hours - 2 weeks)

1. If trade secret misappropriation is confirmed: file for TRO/preliminary injunction within 72 hours
2. If DTSA seizure is warranted: present seizure package to federal judge within 24 hours
3. Preserve all evidence for potential criminal referral (Economic Espionage Act, 18 USC 1831-1832)
4. Notify affected customers, partners, and investors as legally required

### Phase 4: Remediation (Ongoing)

1. Post-incident review: identify root cause, implement corrective measures
2. Update security protocols based on lessons learned
3. Review and update all compartmentalization boundaries
4. Assess whether trade secrets have been lost (i.e., become generally known) and adjust IP strategy accordingly (accelerate patent filings for trade secrets that have been compromised)

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## VIII. TRADE SECRET AUDIT SCHEDULE

| Audit                              | Frequency   | Conducted By             | Scope                                                                                      |
|------------------------------------|-------------|--------------------------|--------------------------------------------------------------------------------------------|
| Access Log Review                  | Monthly     | Rook (Security Director) | Review all access to Compartment 3-5 data; identify anomalies                              |
| Employee Access Audit              | Quarterly   | Rook + COO               | Verify each agent's access level matches their role; remove stale access                   |
| Physical Security Inspection       | Quarterly   | Rook                     | Inspect laboratory, server room, and safe; test access controls                            |
| Supplier Compliance Audit          | Semi-annual | Rook + Legal             | Verify suppliers are complying with confidentiality obligations                            |
| Full IP Audit                      | Annual      | External IP Counsel      | Review entire IP portfolio; update trade secret inventory; verify compartmentalization     |
| Director's Black Book Verification | Annual      | Steele + Rook            | Physical inventory of the Black Book; verify contents match current trade secret inventory |

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## IX. TRADE SECRET INVENTORY — CURRENT HOLDINGS

*Note: This section is intentionally maintained in summary form in this document. The complete trade secret inventory — with specific parameters, values, and conditions — is maintained in Compartment 5 (Director's Black Book) only.*

| Trade Secret ID | Category        | General Description                         | Access Level                   | Document Reference                          |
|-----------------|-----------------|---------------------------------------------|--------------------------------|---------------------------------------------|
| TS-001          | MXene Synthesis | MAX phase etching parameters                | Compartment 4 (Steele + Chen)  | Materials Vol. 2, Restricted Appendix A     |
| TS-002          | MXene Synthesis | Intercalation agent and conditions          | Compartment 4 (Steele + Chen)  | Materials Vol. 2, Restricted Appendix B     |
| TS-003          | MXene Synthesis | Delamination sonication parameters          | Compartment 4 (Steele + Chen)  | Materials Vol. 2, Restricted Appendix C     |
| TS-004          | MXene Synthesis | Size selection centrifugation parameters    | Compartment 4 (Steele + Chen)  | Materials Vol. 3, Restricted Appendix A     |
| TS-005          | CNT Chemistry   | Functionalization reagents and conditions   | Compartment 4 (Steele + Chen)  | Materials Vol. 4, Restricted Appendix A     |
| TS-006          | CNT Chemistry   | Grafting density optimization               | Compartment 4 (Steele + Chen)  | Materials Vol. 4, Restricted Appendix B     |
| TS-007          | CNT Chemistry   | CNT dispersion protocol                     | Compartment 4 (Steele + Chen)  | Materials Vol. 4, Restricted Appendix C     |
| TS-008          | YInMn Blue      | Dopant identity for 10°C above max rated    | Compartment 4 (Steele + Chen)  | Materials Vol. 11, Restricted Appendix A    |
| TS-009          | YInMn Blue      | Dopant concentration for threshold tuning   | Compartment 4 (Steele + Chen)  | Materials Vol. 11, Restricted Appendix B    |
| TS-010          | YInMn Blue      | Solid-state synthesis firing parameters     | Compartment 4 (Steele + Chen)  | Materials Vol. 11, Restricted Appendix C    |
| TS-011          | Manufacturing   | MXene-polymer extrusion temperature profile | Compartment 4 (Steele + Cross) | Manufacturing Vol. 1, Restricted Appendix A |
| TS-012          | Manufacturing   | Extrusion screw design specifications       | Compartment 4 (Steele + Cross) | Manufacturing Vol. 1, Restricted Appendix B |
| TS-013          | Manufacturing   | CVI infiltration parameters (CMC)           | Compartment 4 (Steele + Cross) | Manufacturing Vol. 3, Restricted Appendix A |
| TS-014          | Manufacturing   | Geopolymer mix design                       | Compartment 4 (Steele + Cross) | Manufacturing Vol. 5, Restricted Appendix A |
| TS-015          | Manufacturing   | Ceramic kiln firing schedules               | Compartment 4 (Steele + Cross) | Manufacturing Vol. 3, Restricted Appendix B |

Additional trade secrets are identified and documented in the Restricted Appendices of each relevant Materials Science and Manufacturing volume. The master index of all trade secrets, with cross-references to the specific parameters, is maintained in the Director's Black Book (Compartment 5).

## X. LEGAL FRAMEWORK REFERENCE

| Statute/Act                                          | Relevance                                 | Key Provision                                                                                         |
|------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Defend Trade Secrets Act (DTSA), 18 USC 1836 et seq. | Federal trade secret protection           | Private right of action; ex parte seizure; whistleblower immunity                                     |
| Economic Espionage Act (EEA), 18 USC 1831-1832       | Criminal penalties for trade secret theft | Up to 10 years imprisonment for individuals; fines up to \$5M for organizations                       |
| Uniform Trade Secrets Act (UTSA)                     | State-level trade secret protection       | Adopted by 49 states (all except NY, which follows common law); defines misappropriation and remedies |
| Patent Act, 35 USC 101 et seq.                       | Patent protection framework               | Utility patent term of 20 years from filing date                                                      |
| America Invents Act (AIA)                            | First-inventor-to-file system             | Priority determined by filing date, not invention date                                                |



| Statute/Act                     | Relevance            | Key Provision                                  |
|---------------------------------|----------------------|------------------------------------------------|
| Lanham Act, 15 USC 1051 et seq. | Trademark protection | Federal trademark registration and enforcement |

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